```
========== Priority (Preemptive) ==========
Gantt Chart: [P1 0-5] [P2 5-8] [P3 8-15] [P4 15-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               5       5       0       0
P2      4       3       2               8       4       1       1
P3      6       7       3               15      9       2       2
P4      2       2       4               17      15      13      13

Average waiting time    : 4.00
Average turnaround time : 8.25
Average response time   : 4.00
CPU utilization         : 100.00%
Total idle time         : 0

========== Round Robin ==========
Gantt Chart: [P1 0-2] [P4 2-4] [P1 4-6] [P2 6-8] [P3 8-10] [P1 10-11] [P2 11-12] [P3 12-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               11      11      6       0
P2      4       3       2               12      8       5       2
P3      6       7       3               17      11      4       2
P4      2       2       4               4       2       0       0

Average waiting time    : 3.75
Average turnaround time : 8.00
Average response time   : 1.00
CPU utilization         : 100.00%
Total idle time         : 0
```

```
========== SRTF (Preemptive) ==========
Gantt Chart: [P1 0-2] [P4 2-4] [P1 4-7] [P2 7-10] [P3 10-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               7       7       2       0
P2      4       3       2               10      6       3       3
P3      6       7       3               17      11      4       4
P4      2       2       4               4       2       0       0

Average waiting time    : 2.25
Average turnaround time : 6.50
Average response time   : 1.75
CPU utilization         : 100.00%
Total idle time         : 0

========== Priority (Non-preemptive) ==========
Gantt Chart: [P1 0-5] [P2 5-8] [P3 8-15] [P4 15-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               5       5       0       0
P2      4       3       2               8       4       1       1
P3      6       7       3               15      9       2       2
P4      2       2       4               17      15      13      13

Average waiting time    : 4.00
Average turnaround time : 8.25
Average response time   : 4.00
CPU utilization         : 100.00%
Total idle time         : 0
```

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```
CPU Scheduling Algorithms Simulator
Enter number of processes (1-100): 4
Enter Arrival Time, Burst Time, and Priority for each process.
Use lower priority numbers for higher priority.
P1 (AT BT Priority): 0 5 1
P2 (AT BT Priority): 4 3 2
P3 (AT BT Priority): 6 7 3
P4 (AT BT Priority): 2 2 4
Enter Round Robin time quantum (positive integer): 2

========== FCFS (Non-preemptive) ==========
Gantt Chart: [P1 0-5] [P4 5-7] [P2 7-10] [P3 10-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               5       5       0       0
P2      4       3       2               10      6       3       3
P3      6       7       3               17      11      4       4
P4      2       2       4               7       5       3       3

Average waiting time    : 2.50
Average turnaround time : 6.75
Average response time   : 2.50
CPU utilization         : 100.00%
Total idle time         : 0

========== SJF (Non-preemptive) ==========
Gantt Chart: [P1 0-5] [P4 5-7] [P2 7-10] [P3 10-17]

PID     AT      BT      Priority        CT      TAT     WT      RT
P1      0       5       1               5       5       0       0
P2      4       3       2               10      6       3       3
P3      6       7       3               17      11      4       4
P4      2       2       4               7       5       3       3

Average waiting time    : 2.50
Average turnaround time : 6.75
Average response time   : 2.50
CPU utilization         : 100.00%
Total idle time         : 0
```